

EMLEY MOOR, United Kingdom

WMO: 033460

Lat: 53.612N

Lon: 1.667W

Elev: 267

StdP: 98.16

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.2	-2.2	-5.4	2.5	-0.8	-4.2	2.7	-0.3	28.3	5.6	25.7	4.7	4.3	300	0.690

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.0	23.9	17.5	21.8	16.6	20.2	15.7	18.4	22.2	17.3	20.7	16.3	19.4	4.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.8	12.4	20.1	15.9	11.6	18.8	15.0	11.0	17.8	53.1	22.3	49.6	20.9	46.7	19.4	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.9	18.4	13.9	-4.8	27.0	1.5	2.0	-5.9	28.4	-6.8	29.6	-7.6	30.7	-8.8	32.1		
			-5.1	20.4	1.5	1.3	-6.2	21.3	-7.1	22.0	-7.9	22.8	-9.0	23.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	3.6	3.7	5.2	7.1	10.3	13.1	15.3	15.1	12.8	9.7	6.3	4.3
	DBStd	5.10	2.97	3.15	3.02	2.96	3.02	2.69	2.62	2.39	2.40	2.72	2.81	3.15
	HDD10.0	989	198	177	152	96	32	4	0	0	3	37	113	178
	HDD18.3	3453	456	409	408	338	248	159	100	104	168	267	359	436
	CDD10.0	593	0	1	2	8	42	96	165	159	87	29	3	1
	CDD18.3	14	0	0	0	0	0	2	7	5	1	0	0	0
	CDH23.3	73	0	0	0	0	0	12	35	24	1	1	0	0
	CDH26.7	4	0	0	0	0	0	1	3	1	0	0	0	0
Wind	WSAvg	5.8	7.3	6.9	6.3	5.3	5.2	4.7	4.5	4.8	4.6	6.1	6.4	7.7
Precipitation	PrecAvg	938	88	68	64	67	61	79	72	79	77	92	91	99
	PrecMax	1184	190	156	112	138	115	269	144	192	153	170	180	194
	PrecMin	702	18	10	8	13	16	22	30	22	33	32	46	49
	PrecStd	126	42	34	28	33	24	55	33	42	36	39	40	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.6	13.3	14.2	18.3	22.0	25.1	26.2	26.2	22.6	18.6	13.9	12.7
		MCWB	10.0	10.0	10.0	13.0	16.1	17.5	18.3	18.3	17.4	14.9	12.6	11.4
	2%	DB	10.1	10.7	12.0	15.0	19.5	21.7	24.1	22.9	19.9	16.1	12.7	11.3
		MCWB	8.8	8.7	9.3	10.8	14.2	16.0	17.5	17.4	15.7	13.9	11.5	10.2
	5%	DB	9.0	9.3	10.8	13.1	17.2	19.5	22.2	20.7	17.9	14.7	11.5	9.9
		MCWB	7.7	7.7	8.6	9.8	12.9	14.8	16.8	16.5	14.7	13.0	10.3	8.9
	10%	DB	8.0	8.2	9.5	11.7	15.3	17.8	20.2	19.2	16.5	13.7	10.4	8.9
		MCWB	6.9	6.8	7.7	8.7	11.8	13.8	15.9	15.6	13.8	12.2	9.4	7.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.3	10.6	11.2	13.4	16.6	18.7	19.7	19.6	18.6	15.6	12.8	11.7
		MCDB	11.1	12.1	12.8	17.2	20.4	23.8	24.4	23.7	21.4	17.7	13.6	12.5
	2%	WB	9.0	9.2	10.0	11.5	14.8	16.7	18.3	18.1	16.5	14.3	11.7	10.3
		MCDB	9.9	10.6	11.6	14.3	18.4	20.3	22.5	21.4	18.9	15.8	12.5	11.1
	5%	WB	7.9	8.0	8.9	10.2	13.5	15.5	17.3	17.1	15.1	13.4	10.5	9.2
		MCDB	8.7	9.2	10.3	12.7	16.5	18.7	20.9	20.2	17.1	14.6	11.3	10.0
	10%	WB	7.0	6.9	7.8	9.0	12.2	14.4	16.2	16.2	14.3	12.3	9.5	7.9
		MCDB	7.9	8.1	9.4	11.2	14.8	17.2	19.6	18.7	16.1	13.5	10.3	8.8
Mean Daily Temperature Range	5% DB	MDBR	3.7	4.2	5.1	6.2	6.9	6.8	7.0	6.5	5.8	4.7	3.9	3.7
		MCDBR	5.1	5.2	6.6	8.1	9.9	9.9	10.3	8.9	7.8	5.7	4.9	5.0
		MCWBR	5.0	4.6	4.6	5.0	6.0	5.5	5.0	4.5	4.8	4.2	4.5	4.9
	5% WB	MCDBR	4.9	5.1	6.1	7.7	9.1	8.9	9.2	8.0	6.8	5.6	4.8	5.1
		MCWBR	5.0	4.7	4.8	5.2	5.9	5.7	5.2	4.6	4.8	4.5	4.6	5.1
Clear-Sky Solar Irradiance	taub	0.314	0.328	0.357	0.382	0.384	0.385	0.385	0.372	0.355	0.338	0.313	0.301	
	taud	2.345	2.399	2.322	2.267	2.301	2.318	2.318	2.373	2.430	2.456	2.407	2.315	
	Ebn at Noon	651	772	821	845	864	864	856	847	818	751	651	586	
	Edh at Noon	58	75	99	119	121	120	118	106	89	70	54	49	
All-Sky Solar Radiation	RadAvg	0.65	1.32	2.41	3.74	4.73	4.88	4.73	3.90	2.85	1.61	0.80	0.51	
	RadStd	0.07	0.15	0.28	0.43	0.32	0.46	0.48	0.25	0.25	0.14	0.09	0.05	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (70 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page